Supplementary Material
OnlineNeuro: Active online learning for effective exploration of neural simulation parameters

Contents

1 Sample efficiency in toy problems

A classification example resembling the AP-blocking classification task is presented to evaluate the advantage of active learning with BALD against different search strategies. Four sampling methods are evaluated:

- **Grid search**. A uniform grid is constructed over the parameter space, with resolution proportional to the problem dimensionality. Candidate points are then sampled at random from this grid.
- **Coverage search**. Also referred to as Maximin search, this strategy selects a point from a pool of candidates by maximizing the minimum distance to all previously collected samples. This encourages exploration of under-sampled regions. Formally, this is given by Equation 1, where x is the set of candidate points and y is the set of already collected datapoints.
- **Random search**. At each iteration, a new point is drawn independently from a uniform distribution over the search-space boundaries.
- **Active learning**. Samples are selected according to an acquisition function. For regression tasks, the acquisition rule minimizes global predictive uncertainty (as defined in Equation 2). For classification tasks, we employ the BALD criterion.

$$d_{\min}(\mathbf{x}) = \min_{\mathbf{y}\in\mathcal{S}} \|\mathbf{x}-\mathbf{y}\|, \quad \text{with:} \quad \mathbf{x}_{\text{next}} = \arg\max_{\mathbf{x}\in\mathcal{G}} d_{\min}(\mathbf{x}) \tag{1}$$

$$U(\mathcal{X}) = \int_{\mathcal{X}} \sigma^2(\mathbf{x}')\, d\mathbf{x}' \quad \text{with:} \quad \mathbf{x}_{\text{next}} = \arg\max_{\mathbf{x}\in\mathcal{X}} \Delta U(\mathbf{x}) \tag{2}$$

For each search strategy, surrogate modeling is performed using a Gaussian Process (GP) with a Matérn 5/2 kernel. In addition, for the non-active-learning baselines, we also evaluate a Random Forest model with standard scikit-learn hyperparameters (version 1.6.1), e.g., 100 estimators, a *gini* criterion for splitting, and no depth restriction [1].

The hold-out dataset is a grid of points that are intermediate to the candidate points of the first sampling technique.

Each experiment is initialized with 10 random samples and given a budget of 100 new samples, which are collected one at a time. The initial samples are guaranteed to be balanced. Experiments are repeated 10 times to provide expected performance results.

1.1 N-dimensional hyperspheres

The benchmark consists of a collection of n-dimensional hyperspheres ($N \geq 2$) distributed in a bounded search space. A sample is assigned to the positive class if it lies inside at least one of the hyperspheres; otherwise, it belongs to the negative class. In other words, the decision region is given by the union of all hyperspherical volumes.

Two experimental configurations are considered:

- **2D configuration**. The search space is two-dimensional, with domain boundaries $[-4, 4]^2$. Five non-overlapping circles are placed within this region: one centered at the origin and four centered in the middle of each quadrant (see Figure 1a). The central circle has radius

$r = 1$, while the four surrounding circles have radius $r = 1.5$. Since the circles do not overlap, the total positive-class area is $A_+ = 4\pi(1.5)^2 + \pi(1)^2 = 10\pi$. The total area of the search domain is $A_{total} = 8^2 = 64$. Therefore the exact positive-class proportion is $\frac{A_+}{A_{total}} = \frac{10\pi}{64} \approx 0.49$, so a naive classifier is expected to have about 51% accuracy.

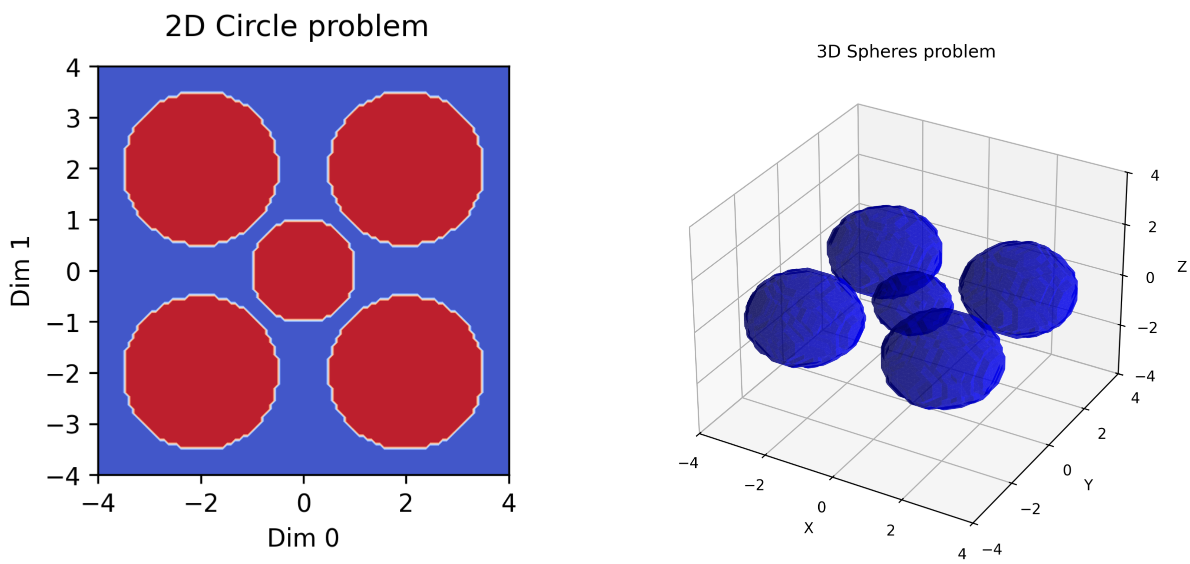
- **3D configuration**. The three-dimensional variant extends the previous setup to the cubic domain $[-4, 4]^3$, corresponding to a total search-space volume of $V_{total} = 8^3 = 512$. Five non-overlapping spheres are placed similarly to the 2D case: one sphere is centered at the origin, while the remaining four are located symmetrically in the surrounding subregions (see Figure 1b). The central sphere has radius $r = 1$, and the four surrounding spheres have radius $r = 1.5$. Since the spheres do not intersect, the total positive-class volume is

$$V_+ = 4 \cdot \frac{4}{3}\pi(1.5)^3 + \frac{4}{3}\pi(1)^3 = \frac{58}{3}\pi$$

and the positive-class proportion is

$$\frac{V_+}{V_{\text{total}}} = \frac{\frac{58}{3}\pi}{512} \approx 0.1186,$$

meaning that only about 11.9% of the domain corresponds to the positive class. As dimensionality increases, the positive region occupies a progressively smaller fraction of the overall search space, resulting in stronger class imbalance.



(a) 2D-classification benchmark.

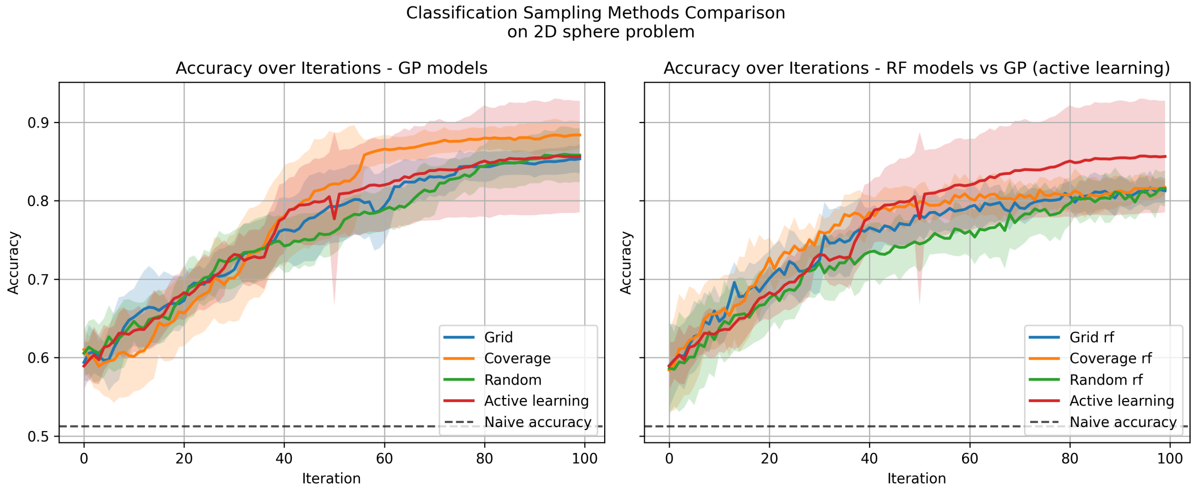
(b) 3D-classification benchmark.

Figure 1: Classification benchmarks.

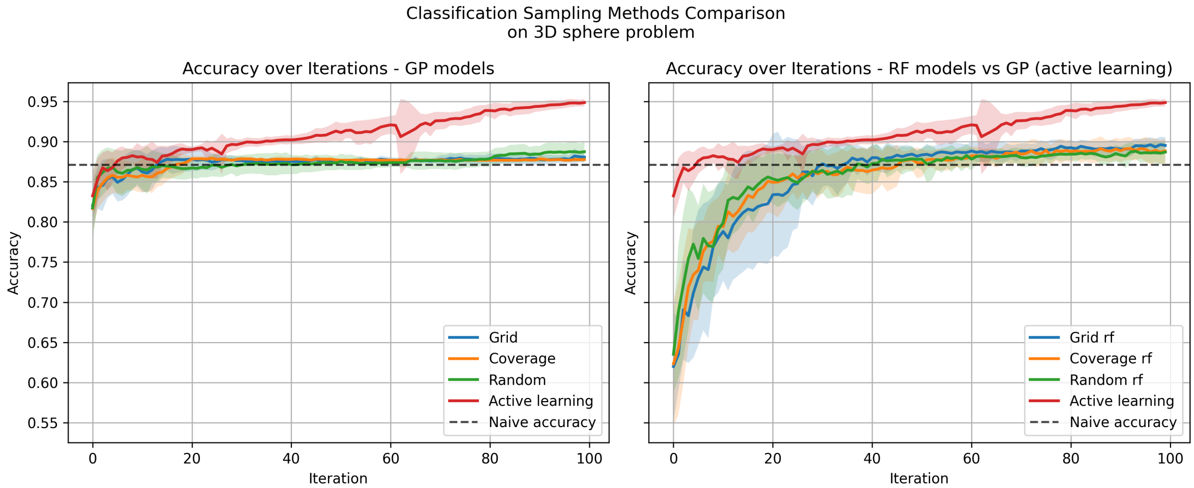
1.2 Results

Figure 2 presents results on hold-out data for each problem. The optimization process uses BALD, which increases accuracy by identifying class boundaries. In the 2D problem, active learning reaches the highest accuracy with fewer samples.

In the 2D scenario, the best sampling method is coverage, as it is a relatively simple problem. However, as shown in the 3D scenario, when the classification problem is higher-dimensional,



(a) 2D classification benchmarking.



(b) 3D classification benchmarking.

Figure 2: Classification benchmark results.

simplistic sampling rules are not capable of outperforming a naive baseline. In addition, results show that GP models are more informative than RF throughout sample collection.

Finally, Figure 3 presents the initial data points and an example of the explored search space for each strategy.

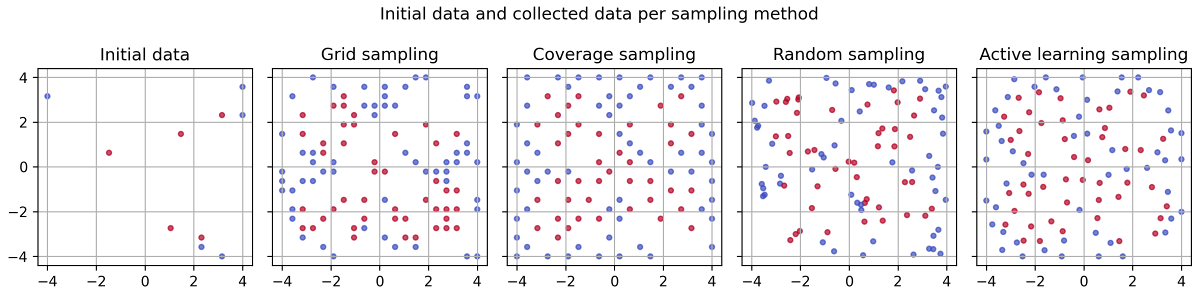


Figure 3: Initial and final search-space exploration for each search strategy for the 2D problem.

2 Temporal Variance–Movement Sampling

Given a probabilistic model with predictive mean $\mu(x)$ and variance $\sigma^2(x)$, we define a sampling score that combines local sensitivity of the mean prediction with predictive uncertainty.

For a candidate point x, the score is defined as:

$$S(x) = \alpha\,\tilde{g}(x) + (1-\alpha)\,\tilde{\sigma}^2(x), \tag{3}$$

where $\alpha \in [0, 1]$ is a weighting factor.

The sensitivity term $g(x)$ is the norm of the gradient of the predictive mean,

$$g(x) = \|\nabla_x \mu(x)\|_2\,, \tag{4}$$

and $\sigma^2(x)$ denotes predictive variance.

Both terms are normalized across the candidate set using min–max scaling,

$$\tilde{f}(x) = \frac{f(x) - \min f}{\max f - \min f}, \tag{5}$$

ensuring comparable magnitudes before combining them.

This emphasizes segments with increased temporal changes, which in this context emphasizes correct prediction of action-potential peaks.

References

[1] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.